Performance V2: measured deltas

Qidian (Martin) Xing | 78,682 firm-years, 8,971 firms; 66,400 records through 2014; 12,282 previously inspected 2015-2018 records.

Recipe	Cutoff	Input	Sel	Seeds	Yr AP	Pool AP	ROC	R@5	Min	Source
V0 base	2011	18/01	Y	1	NA	0.145	0.835	NA	NA	Archive
V0 L1	2011	13/01	N	1	NA	0.295	0.858	NA	NA	Archive
V1 RF	2012	S1	Y	1	0.215	0.206	0.862	0.460	0.428	V1
V1 DL	2012	S1	Y	1	0.110	0.105	0.815	0.294	0.602	V1
P2 old L1	2014	L1/01	Pre	5	0.262	0.259	0.862	0.524	2.798	Mean score
P2 lag XGB	2014	Lag3/V1	Pre	5	0.303	0.296	0.853	0.513	0.605	Mean score
P2 Cat raw	2014	RAW/V1	Pre	5	0.279	0.271	0.852	0.488	0.583	Mean score
P2 V1 RF	2014	S1/V1	Pre	5	0.286	0.286	0.891	0.504	1.399	Mean score
P2 V1 DL	2014	S1/V1	Pre	5	0.228	0.214	0.874	0.462	1.057	Mean score
P2 overall	2014	LEGACY33/native/half-life3	Y	5	0.313	0.325	0.904	0.540	0.331	Mean score
P2 DL	2014	RAW17/01/half-life3	Y	5	0.393	0.392	0.912	0.621	5.487	Mean score
P2 blend	2014	annual ranks	Y	5	0.373	0.383	0.922	0.620	NA	Rank blend
Rank old L1	2014	annual ranks	Pre	5	0.262	0.254	0.861	0.524	NA	Rank reference
Rank lag XGB	2014	annual ranks	Pre	5	0.303	0.300	0.853	0.513	NA	Rank reference
Rank Cat raw	2014	annual ranks	Pre	5	0.279	0.270	0.851	0.488	NA	Rank reference
Rank V1 RF	2014	annual ranks	Pre	5	0.286	0.281	0.892	0.504	NA	Rank reference

The development-selected fixed annual-rank blend has historical annual AP 0.373, versus 0.303 for the strongest frozen same-cutoff baseline point estimate (baseline_v1_lags_xgb). The difference is +0.070. It exceeds all four annual-AP baseline point estimates. All four paired annual-AP lower bounds are positive under the conditional company bootstrap.

Yr AP is equal-weight annual AP; Pool AP and ROC pool years; R@5 averages annual review recall. Min is measured fit time. Mean-score ensembles differ from mean seed metrics. Sel is selection within each version/family. Archive rows are source observations, not new experiments. Annual ROC guardrails and full identities are in selection_trace.csv and baseline_bridge.csv.

2. Why the legacy recipe could look stronger

2012 controlled recipe	AP	Seconds
xgb_raw_keepbest	0.239	0.654
legacy192_s1	0.206	22.082
legacy192_raw18	0.249	59.504
xgb_LAGS_RAW3	0.333	6.668
legacy192_inner_l1	0.214	170.366
xgb_RATIO_CORE	0.291	6.149
xgb_RAW17	0.285	1.123
legacy192_epochrefit	0.256	62.872
legacy192_raw17	0.325	196.214
legacy192_clip005	0.314	157.399
legacy192_quantile	0.243	26.896
legacy64_raw18	0.259	18.227
legacy192_keepbest	0.263	37.591
legacy192_unweighted	0.364	117.674
legacy192_ros_steps	0.392	141.263
legacy192_ros_epochs	0.417	234.856

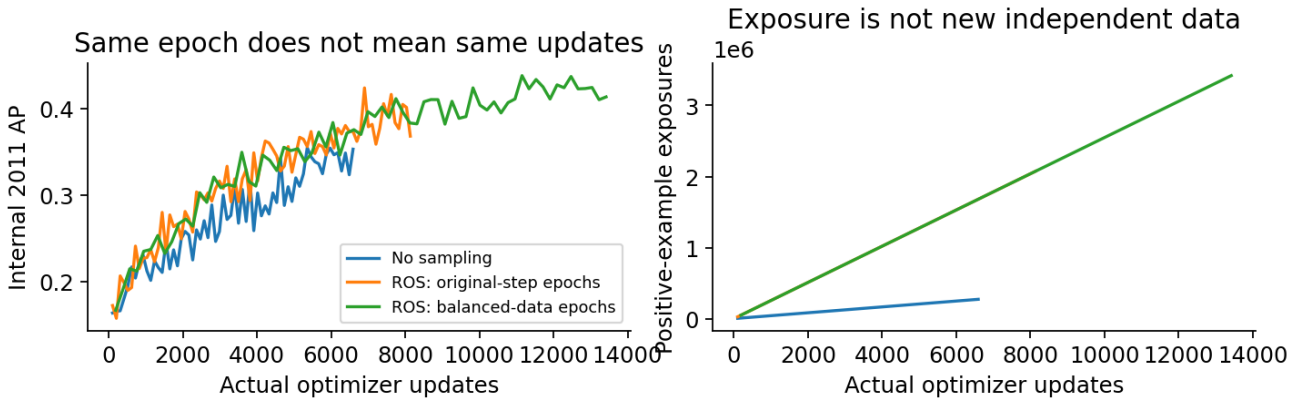
The legacy residual network has width 192/four blocks and about 302k parameters, versus 36,801 for the V1 selected width 64/two-block ROS pipeline. The latter used 18 final epochs; the V1 core table used a different 16-epoch configuration. Inputs, clipping, loss weights, stopping information and refitting also changed.

Legacy and sklearn AP agree on actual saved predictions and tied-score fixtures. Capacity differences therefore establish a changed experiment, not an automatic explanation of generalization. These seed 42/single-year controls provide conditional mechanism evidence, not universal causal attribution. The included epoch 31 checkpoint is distinct from the bridge epoch 23 record; exact archived 13 selection never enters early P2 development folds.

In the fixed 2012 control, reducing RAW18 capacity from 192 x 4 to 64 x 2 changes AP from 0.249 to 0.259; this does not support bigger-is-always-better. Removing only the duplicate raw column gives 0.325; removing full class weighting gives 0.364. These are local mechanism sensitivities, not a causal decomposition of the historical version gap.

Actual archive-role results: original-loop fixed-13 retraining has pooled AP 0.225; the included epoch 31 checkpoint gives 0.140. The former does not reproduce archived 0.295; the latter is not the archived epoch 23 artifact. An archive-cache row permutation was corrected for these two rows, then every frozen pipeline was reloaded in source-row order, without new fitting or selection changes.

3. Training budgets and numerical representation



Actual updates and positive exposures distinguish optimization budget from sampling.

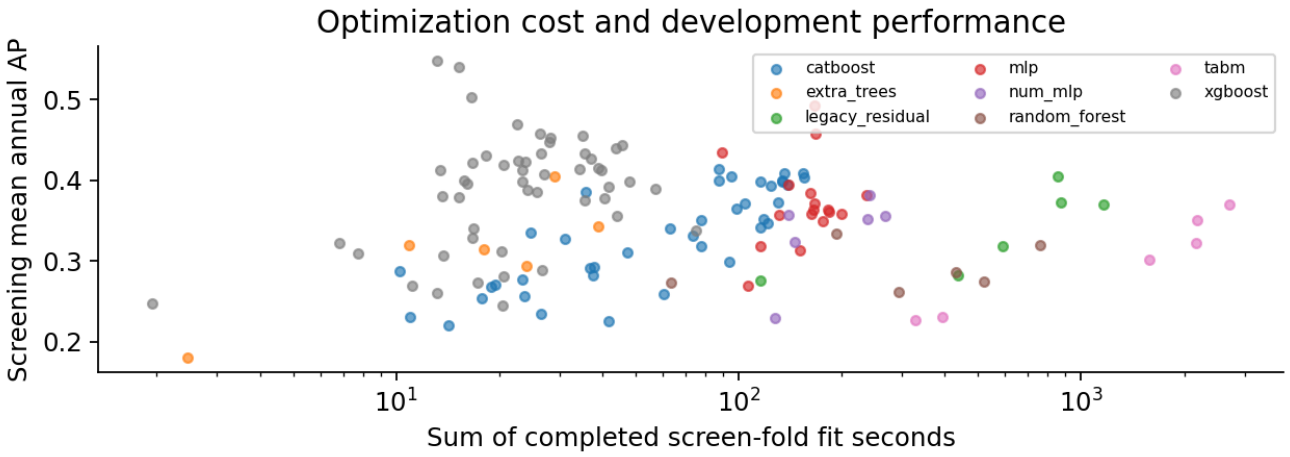
Fixed-update control	Updates	Positive exposures	Train BCE
bridge_equal2000_none__seed42__year2012	2000	80778	0.246
bridge_equal2000_smote__seed42__year2012	2000	509988	0.498
bridge_equal2000_ros__seed42__year2012	2000	509988	0.480

Three treatments receive 2,000 actual updates. Numerical transforms and bin knots fit current training roles only; binary masks bypass quantile mapping. Constant/duplicate masks are removed without dropping records. Official numerical embedding modules and TabM are used; TabM averages per-head losses during training and per-head sigmoid probabilities at inference.

A V2 migration error in the V1 ROS epoch-equivalent refit used the natural row count. That trial was invalidated, retained and replaced before selection. It was not a defect in preserved V1 code, and ordinary P2 step-refit HPO was unaffected.

4. Search coverage, selection and confirmation

Family	Complete	Failed	Pruned
catboost	36	0	0
xgboost	36	0	0
random_forest	6	0	0
extra_trees	6	0	0
legacy_residual	6	0	0
mlp	6	0	0
num_mlp	6	0	0
tabm	6	0	0
official_ft	0	0	0
xgb_rank	0	0	0



Real three-year screening configurations; folds/seeds are not additional methods.

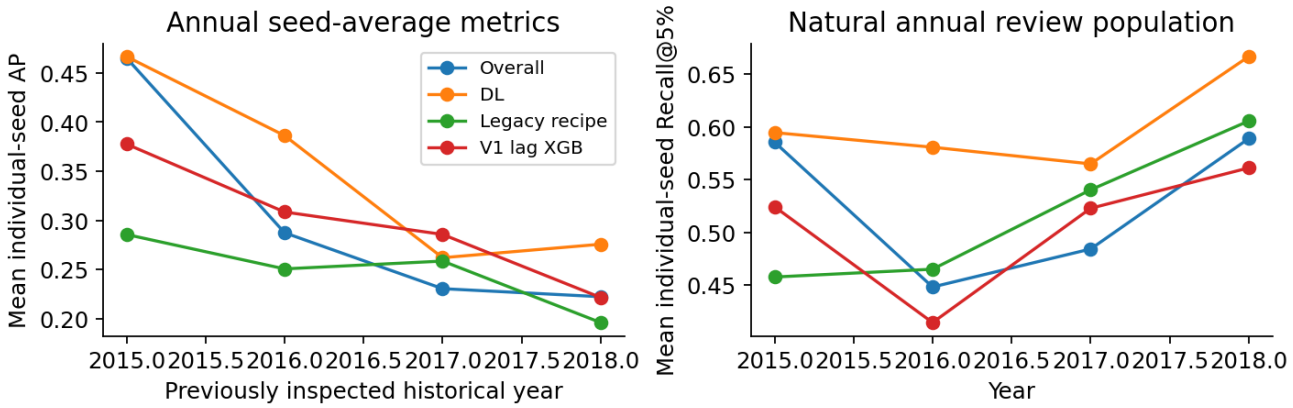
Screening years are 2009/2011/2013. Shortlisted candidates complete all five 2009-2013 years and three registered seeds. Selection averages seed/year AP, with 0.005 near-tie tolerance, recent-year/review-capacity/cost tie-breaks and ROC/Top5 guardrails. Strong fixed baselines remain eligible. Candidate/blend choices freeze before one 2014 confirmation batch; confirmation cannot reopen search.

5. Historical evidence against strong baselines

The development-selected fixed annual-rank blend has historical annual AP 0.373, versus 0.303 for the strongest frozen same-cutoff baseline point estimate (baseline_v1_lags_xgb). The difference is +0.070. It exceeds all four annual-AP baseline point estimates. All four paired annual-AP lower bounds are positive under the conditional company bootstrap.

Descriptively, the overall single recipe's raw pooled AP is 0.325, versus archived legacy L1 0.295 (+0.029). Different cutoff/selection/seed roles prevent attributing this archive difference to the algorithm; it is separate from the same-cutoff annual-AP comparison.

Same-cutoff reference	Annual AP delta low	High	Draws
baseline_legacy_strong_inner_l1	0.039	0.186	2000
baseline_v1_lags_xgb	0.002	0.144	2000
baseline_v1_raw_catboost	0.029	0.166	2000
baseline_v1_selected_rf	0.032	0.148	2000



Averages of individual-seed metrics, distinct from mean-score ensemble metrics.

The archived L1 pooled AP 0.295332 came from one seed and different selection/training roles and was not the old validation-ROC-selected representative. P2 retrains every comparison baseline through 2014. P2-selected cutoff 2012 transfers are explicitly separate from strict P1. Annual-rank pooled AP is compared only with matching annual-rank baselines.

What changed: measured recipe and confirmation

overall selected recipe: {"depth": 8, "family": "xgboost", "features": "LEGACY33", "half_life": 3, "iterations": 1500, "learning_rate": 0.07016184471619624, "preprocess": "native", "sampling": "none", "weight_alpha": 0}; five-year/three-seed AP=0.529.

DL selected recipe: {"depth": 2, "family": "mlp", "features": "RAW17", "half_life": 3, "learning_rate": 0.001, "preprocess": "std01", "sampling": "none", "schedule": "fixed", "weight_alpha": 0, "width": 128}; five-year/three-seed AP=0.500.

Backbone	Input/scale	No embedding AP	PLE AP	Change
num_mlp	RAW/01	0.382	0.357	-0.025
num_mlp	Log/QT	0.324	0.356	0.033
tabm	RAW/01	0.323	0.351	0.029
tabm	Log/QT	0.231	0.301	0.070

One 2014 confirmation	AP	ROC	R@5
Overall	0.572	0.906	0.592
DL	0.527	0.905	0.556
legacy_strong_inner_l1	0.311	0.823	0.430
v1_lags_xgb	0.311	0.822	0.401
v1_selected_rf	0.422	0.877	0.493

Neural single recipe: selected DL annual AP=0.393; same-cutoff V1 DL=0.228 (difference +0.165), legacy L1=0.262 (difference +0.131). These are historical point estimates from mean five-seed predictions; stability and intervals are reported separately.

PLE deltas are matched-backbone, one-seed, three-year development results. Multivariable final changes support pipeline comparisons; the 2014 batch never triggers reselection.

Fixed selected recipe and seed 42, changed fit cutoff: annual AP=0.242 through 2012 versus 0.305 through 2014. This P2-selected transfer used 2013 selection information and is not strict P1; rolling recipes shift the start with the same window-length rule.

Compute disclosure: the first eight-hour window expired with partial development evidence. The owner explicitly authorized a separately registered additional eight-hour continuation. Both clocks and prior partial outputs are preserved; final results use additional authorized compute. Actual timing and fit calls are in experiment_summary and fit_accounting.

Three-year development input controls under the same tree recipe: RAW17=0.400, RATIO_CORE=0.322, LAGS_RAW3=0.392. These seed 42 conditional controls retain exact configs/deltas in matched_input_controls.csv.

Development-only one-factor controls: xgboost half_life=3: 0.458 -> 0.548; mlp half_life=3: 0.382 -> 0.493. A three-year half-life halves relative weight at three years of age, normalized to mean one; evaluation records are unchanged. Economic explanations require event dates, and historical pipeline gains are not isolated causal effects.

6. Cost, uncertainty and reproducibility

Recipe	Seeds	Mean annual AP	Min	Max
baseline_legacy_strong_inner_l1	5	0.248	0.225	0.300
baseline_logistic	1	0.169	0.169	0.169
baseline_v1_lags_xgb	5	0.298	0.294	0.304
baseline_v1_raw_catboost	5	0.274	0.262	0.284
baseline_v1_selected_dl_corrected_step	5	0.181	0.161	0.211
baseline_v1_selected_rf	5	0.281	0.273	0.295
refine_r3_15_a028aa7a	5	0.301	0.293	0.316
refine_r3_21_bbf84fb1	5	0.348	0.327	0.388

Paired company bootstrap uses 2,000 draws and carries all annual records together. Intervals condition on fitted models and the current historical sample; they omit full HPO/training uncertainty and common macro shocks. Rank comparisons additionally fix original cohort maps. The company holdouts are accurately labeled post-selection stress, and 2014-cutoff naturally unseen failed firms are few.

All trials, invalidated/failed attempts, prediction coverage, final artifacts and numerical checks are retained. The local model directory contains preprocessing, feature order, states, configs and fit histories; raw data, complete predictions and private cases are excluded from public Git. Reload/prediction commands and exact dependencies are in README and requirements_v2.txt.

This is an owner-directed, Codex-assisted independent extension of an earlier UCLA MFE 413 team project. Parent acknowledgements remain intact. Company-constant labels, missing event/publication dates and prior historical exposure preclude verified one-year PD or prospective deployment claims. Official primary implementation sources and versions are documented; no paper benchmark performance is assumed.

The DL representative has historical annual AP 0.393, above the development-frozen blend at 0.373. Fusion did not improve historical DL AP. The frozen choice is retained rather than replacing it with the highest historical row.

Role	Groups	AP mean	Min	Max	R@5
DL	3	0.138	0.077	0.206	0.413
ML	3	0.143	0.097	0.206	0.360

The common pre-2015 population leaves 468 naturally unseen companies, including 5 failed firms. Actual rolling-window fitting sets were registered separately before final fitting. The table reports mean/range of pooled AP over three company groups, not confidence intervals; hyperparameters were selected on broader companies.